

Wearable BLE Heart Monitor PIFA Antenna

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Assumptions and Heart Monitor Considerations

- We are designing a PIFA antenna to transmit data from a wearable heart monitor using BLE.
- We are assuming that there is a strong input signal which can be efficiently transmitted by the PIFA antenna.
- We are assuming the antenna will be mounted on the wearable device itself rather than directly on the human body, so factors due to skin and the rest of the body were not considered.
- We are assuming the wearable itself will have minimal impact on antenna performance.
- The PIFA antenna we are using will have a ground plane and shield the body from most of the radiation. Additionally, BLE is intentionally lower power so there shouldn't be any safety concerns due to this antenna.

Bluetooth Low Energy (BLE) Basics

- BLE allows for data transfer similar to Bluetooth while using less power. This is accomplished by sending data in short bursts rather than constantly keeping communication open like normal Bluetooth.
- BLE is ideal for this project's use case as a wearable heart monitor should be light and not have a large battery, so minimal power consumption is ideal. Additionally, for heart monitoring constant communication is not necessary, the heart data can be sent in packets instead of constantly, so use of BLE is logical.



BLE Protocol

- Power density limit is 10 by the FCC 40 by the ICNIRP.
- Peak SAR limit is 1.6 by the FCC and 2 by the ICNIRP.
- RF identification (RFID) and near-field communication (NFC) inherently combine wireless communication and power delivery within a single system.
- BLE Protocols:
 - Data Rate: Low-Moderate
 - Communication Range: Moderate
 - Power Consumption: Moderate
- BLE Data Transfer Technique:
 - Frequency Band: 2.4 GHz
 - 40 2 MHz channels
 - Data Rate: <2 Mbps
 - Communication Range: <400 m
 - Power Consumption: <20 mA
 - Technical Standard: IEEE 802.15.1



Bluetooth Low Energy (BLE)

- BLE radio devices conserve power by turning on when necessary.
- With only three channels available to establish a connection between devices, BLE consumes less energy and connects faster.
- Primary advertising channels are used to find devices/send signals.
- BLE devices have a longer battery lifespan due to the use of periodic short bursts to transmit data and battery recovery during idle periods.
- BLE Protocol uses an asymmetric design:
 - One central device that uses more power and has more processing capability.
 - Peripheral that has less capabilities and limited energy resources.



BLE Protocol Stack



Physical Layer (PHY)

Outlines requirements for hardware (LE radio) used in data transmission and reception over the air.



Link Layer

In charge of setting up LE radio to create communication links according to the kind of communication.



Host Controller Interface (HCI) Layer

Serves as an interface between host's higher-level protocols and controller's lower-level protocols.



Logical Link Control & Adaptation Protocol Layer (L2CAP)

Facilitates communication between BLE stack's top and bottom layers.

BLE Protocol Stack



Security Manager (SM)

Outlines how LE devices are paired, authenticated, and encrypted.

Guards against tracking, Man-in-the-Middle attacks, and passive eavesdropping.



Attribute Protocol (ATT)

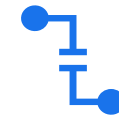
Data type known as an attribute is used by LE devices to store data.

An attribute's structure and access permissions (read, write, read-only, etc.) are specified by the ATT.



Generic Attribute Profile (GATT)

Offers a file-folder-based hierarchical data structure.



Generic Access Profile (GAP)

Outlines how a BLE device can connect or broadcast to other devices.

Design of Wireless-Enabled Wearable Devices

Faster data transfer and broader coverage come at the cost of power consumption.

Applications that require frequent data transmission require rapid low-latency communication.

Devices that only transmit data periodically can function with slower speeds.

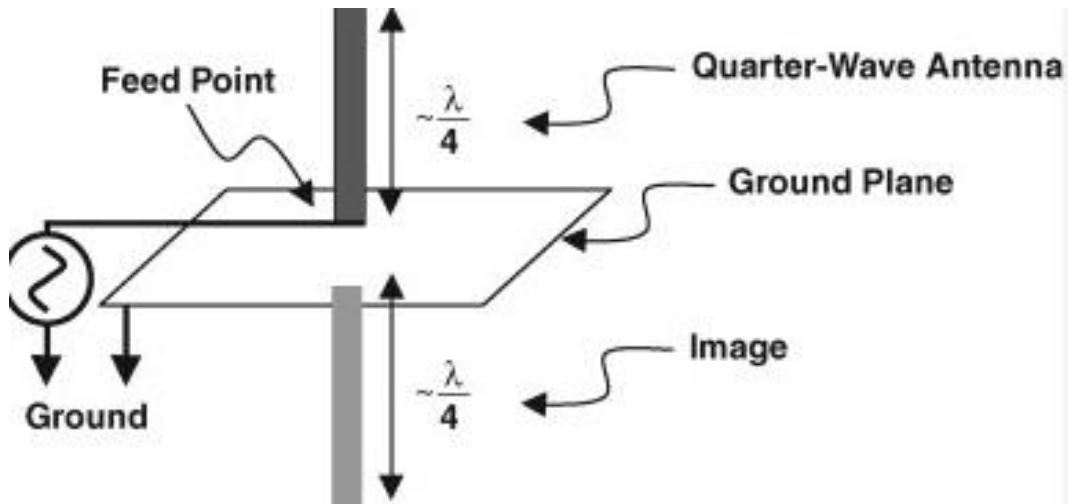
Long-range communication provides greater mobility but requires more energy.

Many wearable electronics communicate primarily with nearby devices.

Overlap of frequency bands causes interference.

Results in distorted signals, data loss, or increased latency

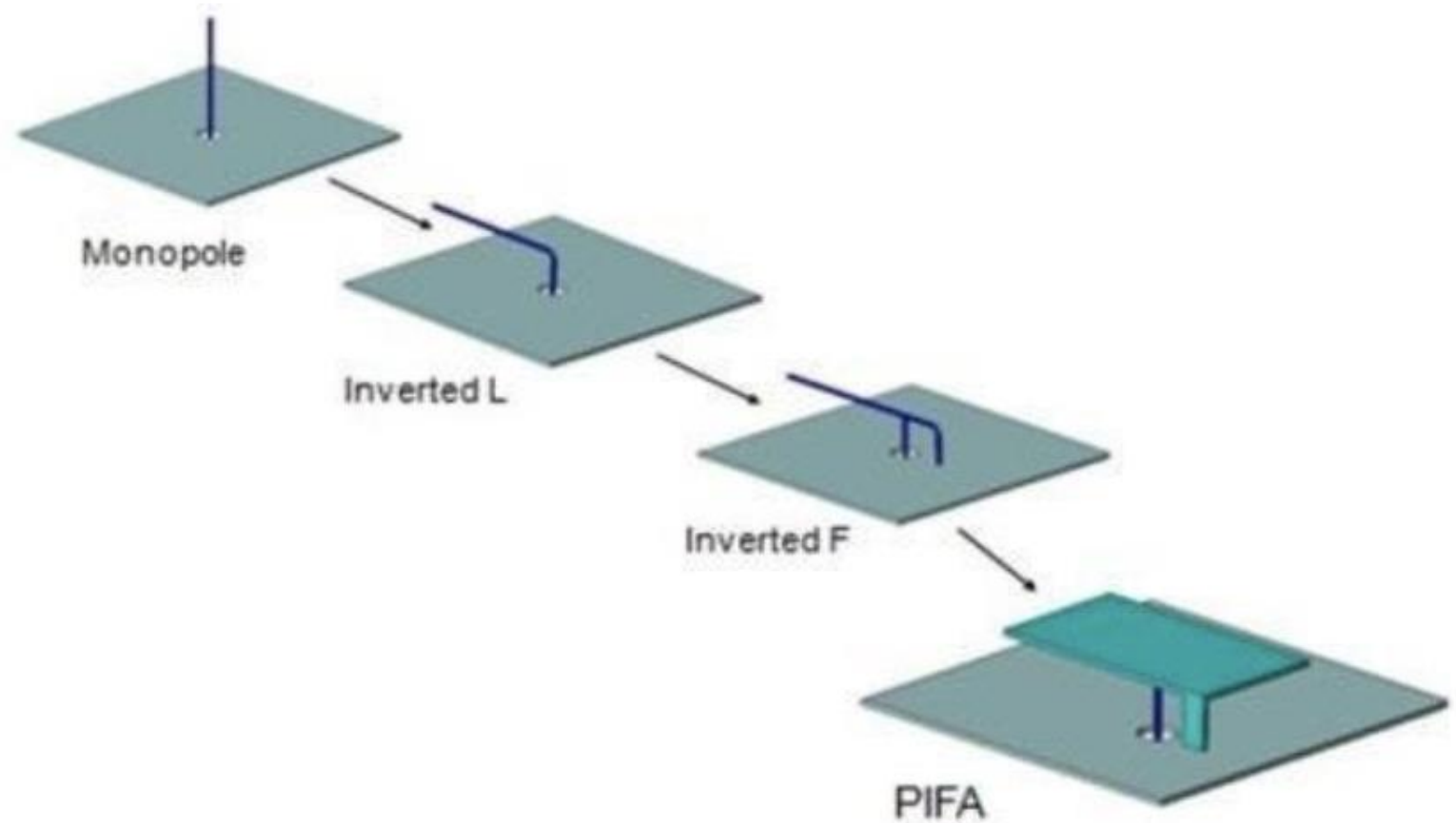
Monopole Antennas



- Monopoles can be seen as half of a dipole that sits above the ground plane.
- The current distribution is like that of a dipole, while the terminal voltage is about half.
- The directivity of the monopole is approximately twice that of a dipole.
- While monopoles require a $\lambda/4$ electrical length, their physical size is often too big for compact devices
- PIFAs are a type of monopole that can be implemented for smaller wireless systems and wearables.

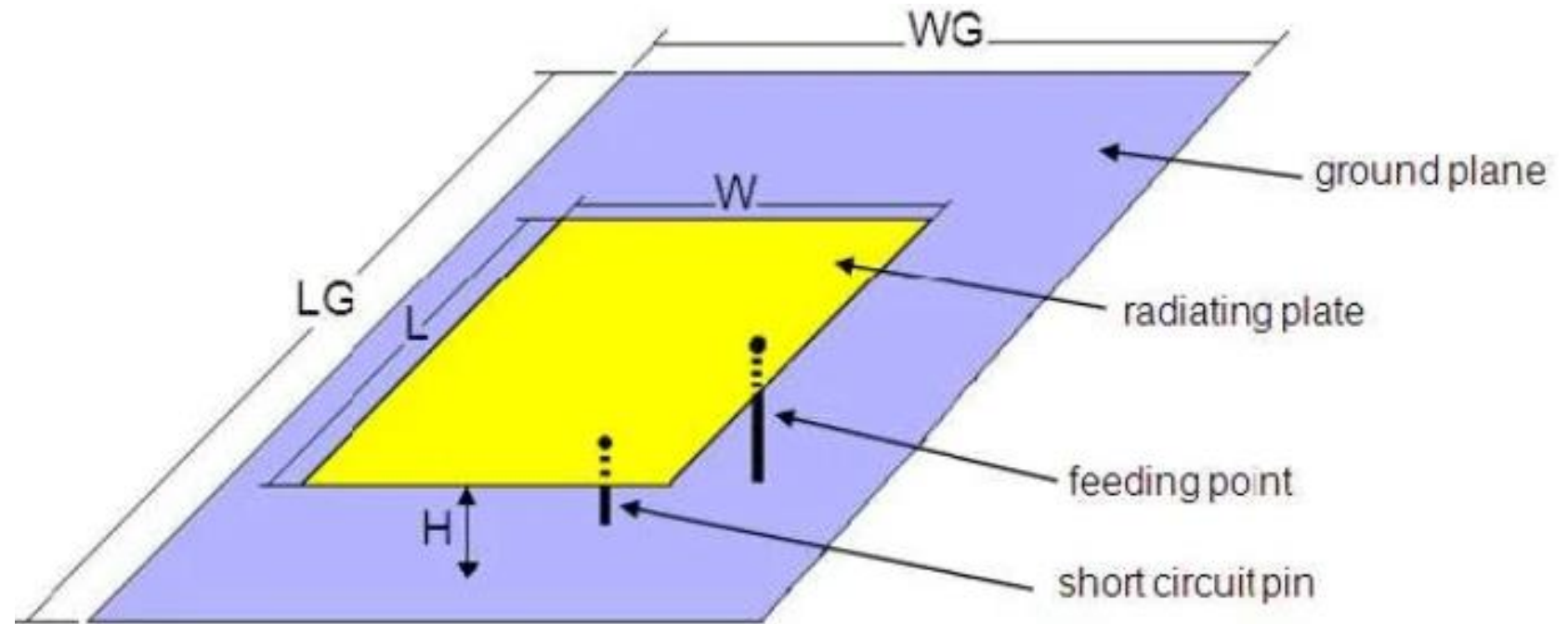
PIFA (Planar Inverted F Antenna) Basics

- Compact planar monopole antenna.
- Formed by folding the quarter wavelength monopole.
- Shaped like an "F" due to the feed and shorting pin.
- Common applications include phones, Bluetooth devices, and Wi-fi routers.



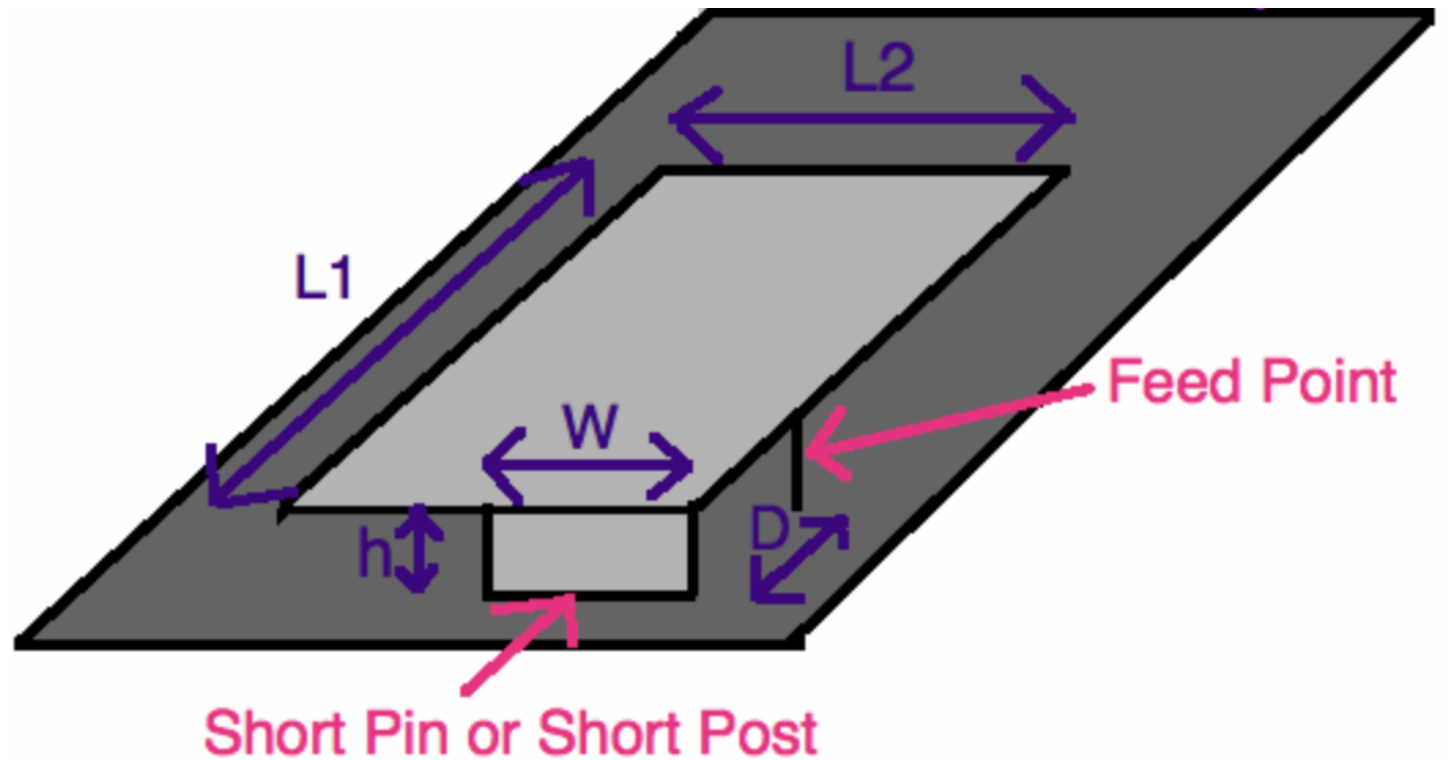
PIFA Structure

- Radiating patch on top
- Ground plane that acts as reference
- Shortening pins connecting to ground
- A Feed point
- Substrate



PIFA Operation

- Electrical Length is approximately a quarter wavelength
 - $L1 + L2 = \lambda/4$
- Feed- short distance controls impedance
 - Closer → Lower
 - Further → Higher
- Naturally close to 50 Ohms matching
- No external matching needed usually
- Radiation pattern is close to omnidirectional



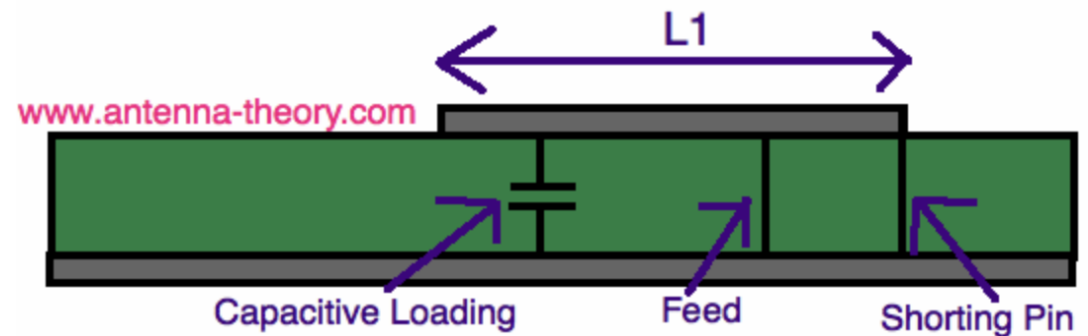
How Different PIFA Dimensions Affect BW / Frequency

Increasing electrical length lowers resonant frequency, while increased spacing improves bandwidth.

Dimension	BW	Frequency
H	Increases	Decreases
W	Increases	Increases
L1	Increases Slightly	Decreases
Feed Plate Width	Increases	Increases
L2	Increases Slightly	Decreases
D	Increases	Decreases Slightly

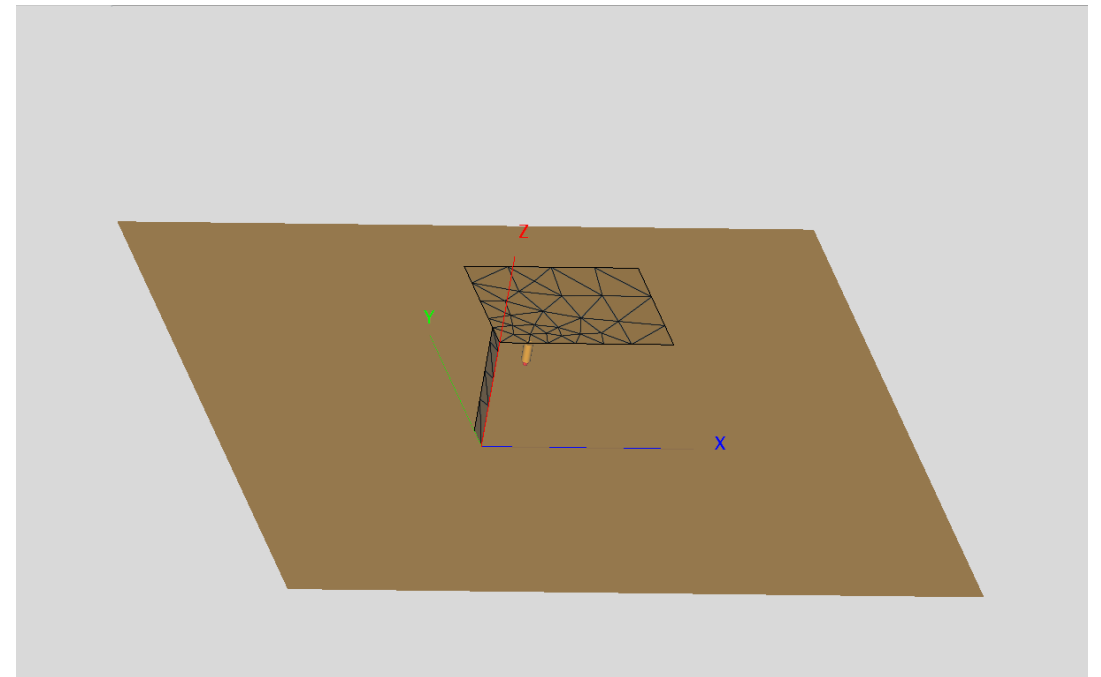
Capacitive loading to reduce PIFA size

- Add a capacitor from the plate to ground
- Increases the antenna's electrical length
- Lowers the resonant frequency without changing the size
- Allows for the PIFA to be made smaller for the same bandwidth



Initial Design

- Infinite PEC ground-plane
- Mismatched L2 and W
- Feed port with a wire on the right edge
- H quite tall and arbitrary



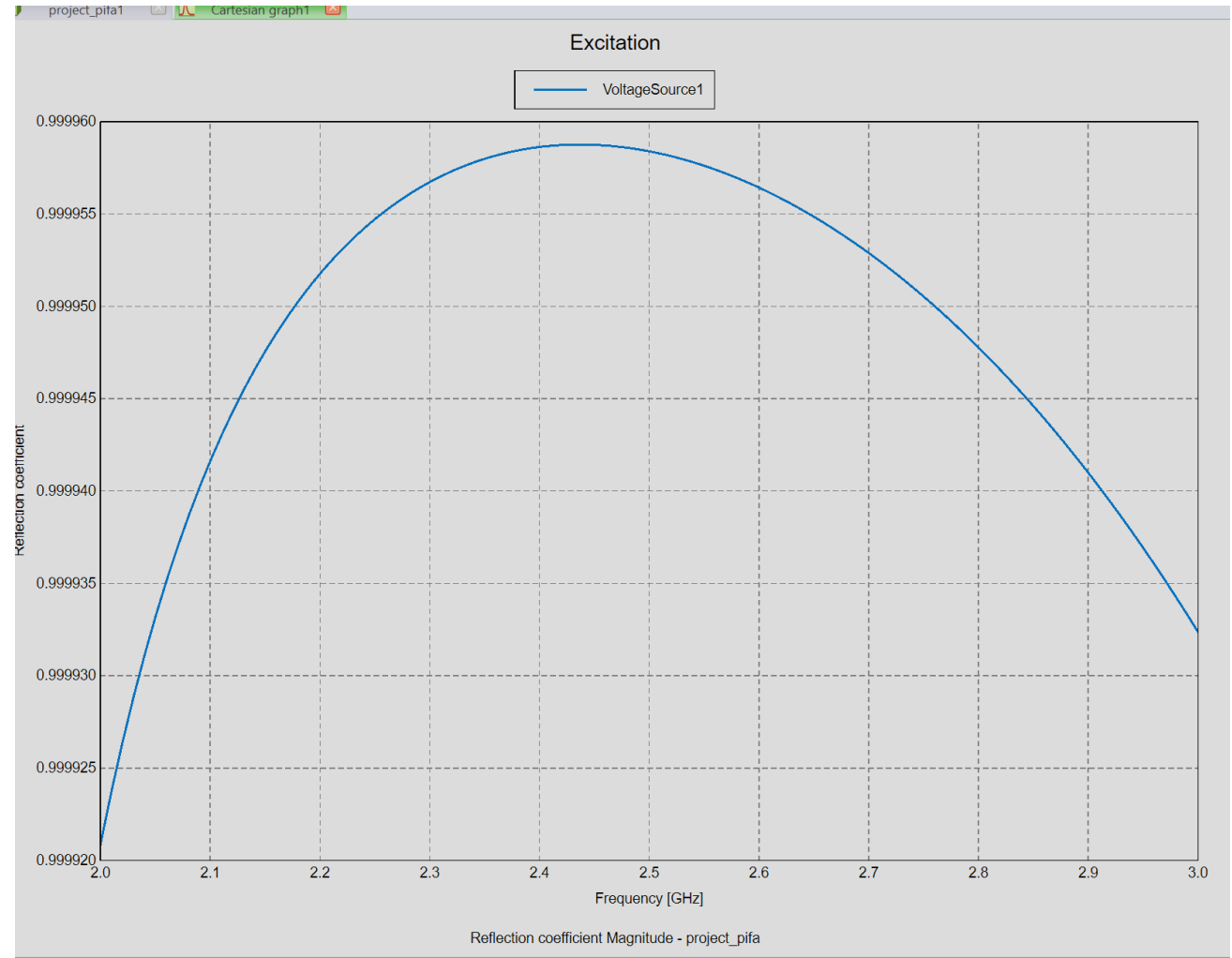
Initial Sim Results- Reflection Coefficient

Matching was quite bad

Tried for many combinations of L1/L2

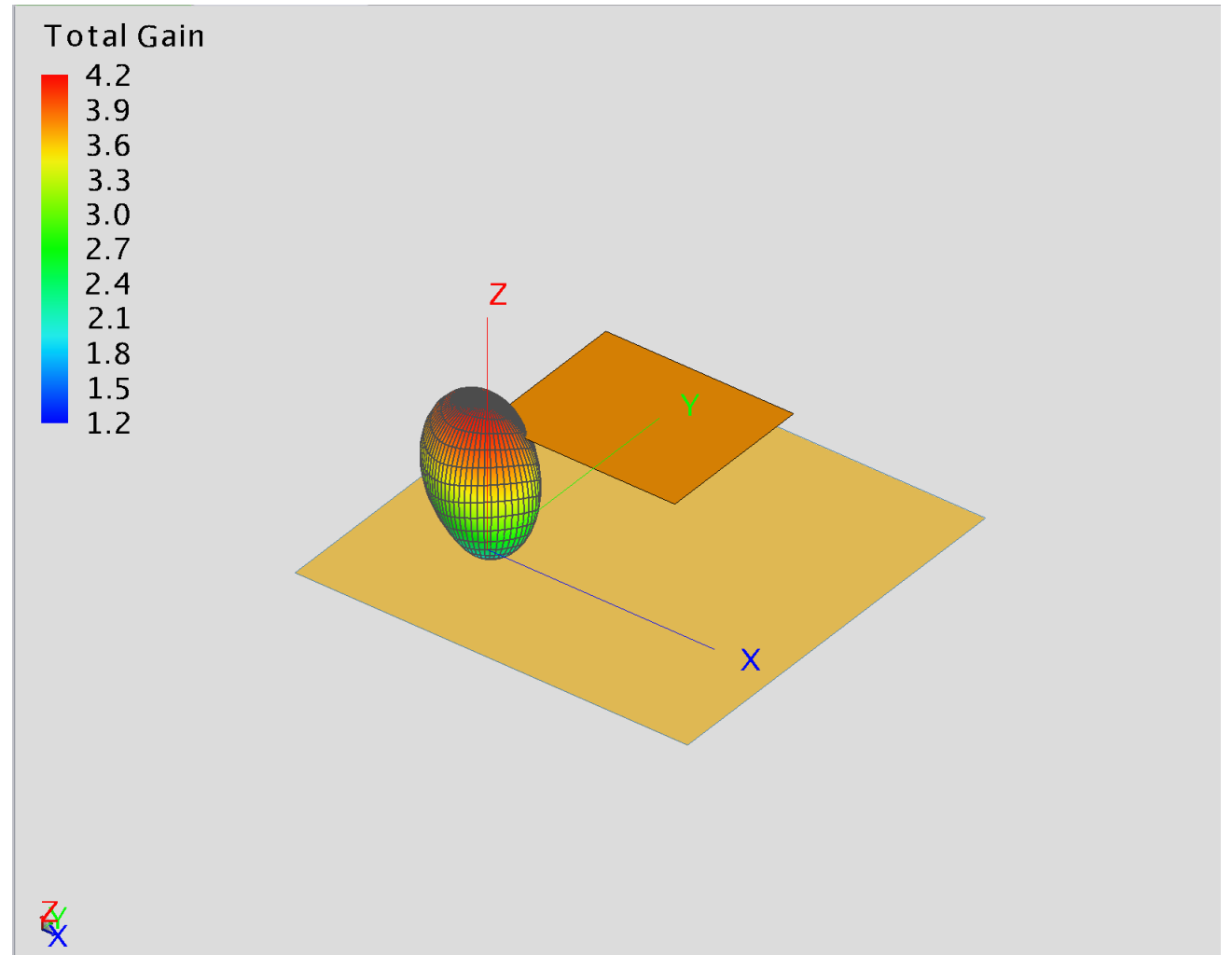
Tried different feed distancing

Still a very bad design



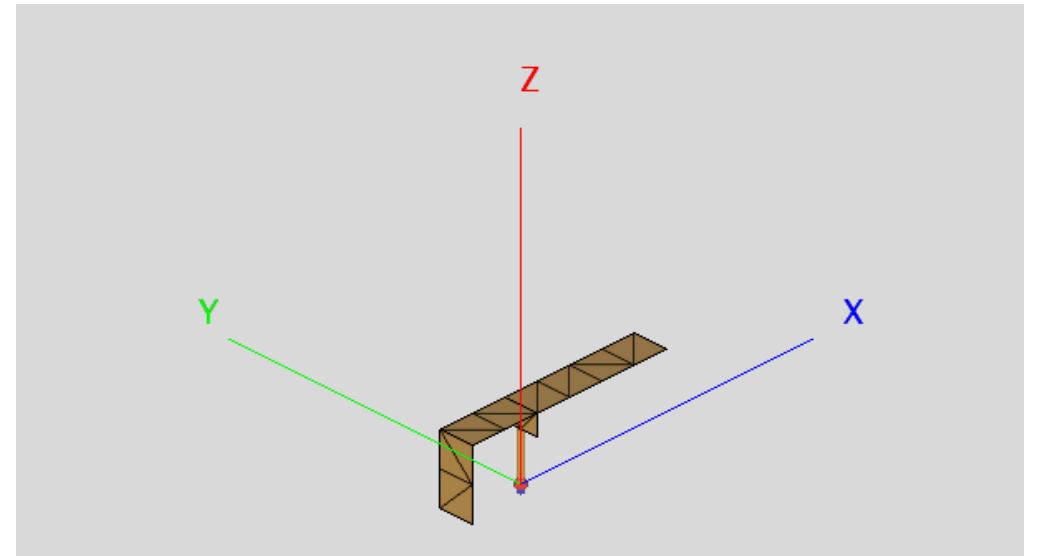
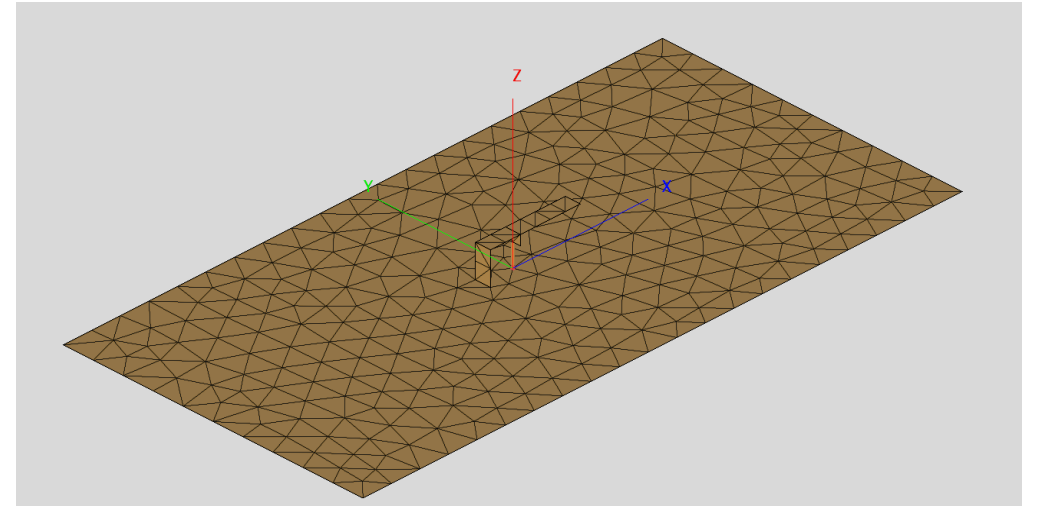
Initial Sim Results- 3D Gain Pattern

- Pattern was very directive when should be approximately isotropic
- Gain was low even at maximum and quite low outside of the peak.



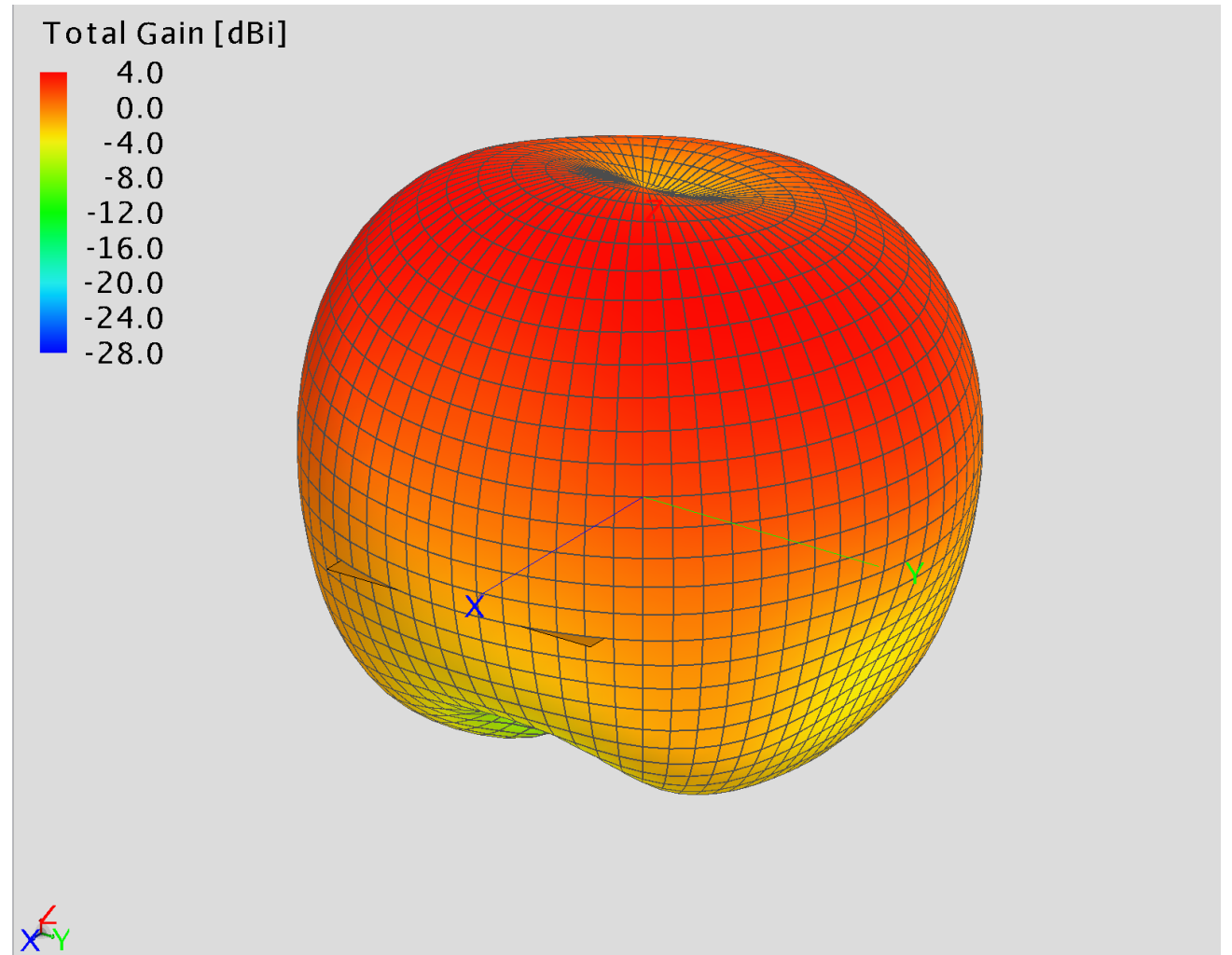
Second Design

- Finite(but large) PEC ground plane
- Matched $L2$ and W
- Fed by W width strip with wire, centered along the antenna
- Starting point for design was $L1 + h = \lambda/4$



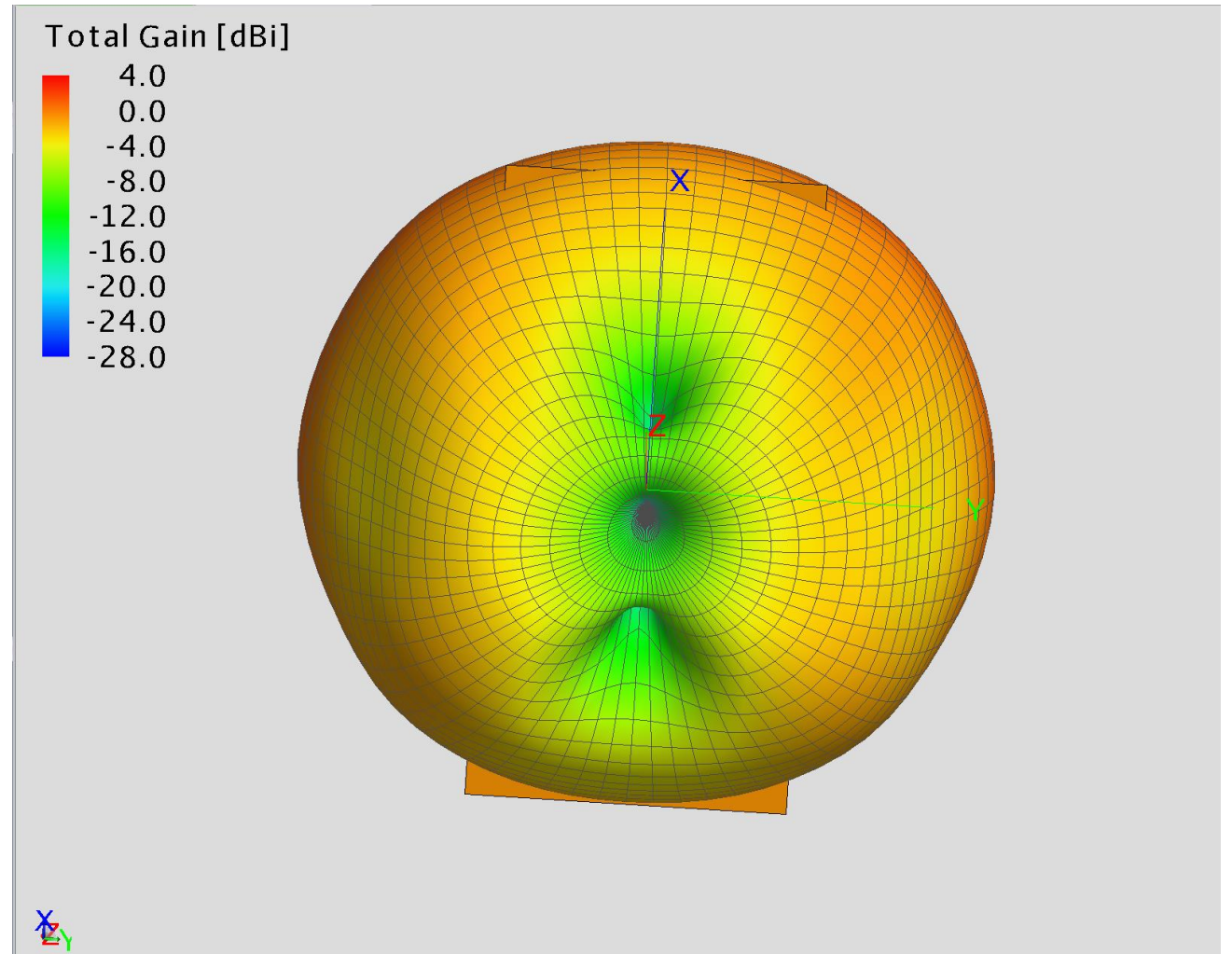
Final Sim Results – 3D Gain Pattern

- Pattern is quite isotropic, especially in the directions we care about for this application.
- Much lower gain towards chest, which is desirable both for safety and performance.



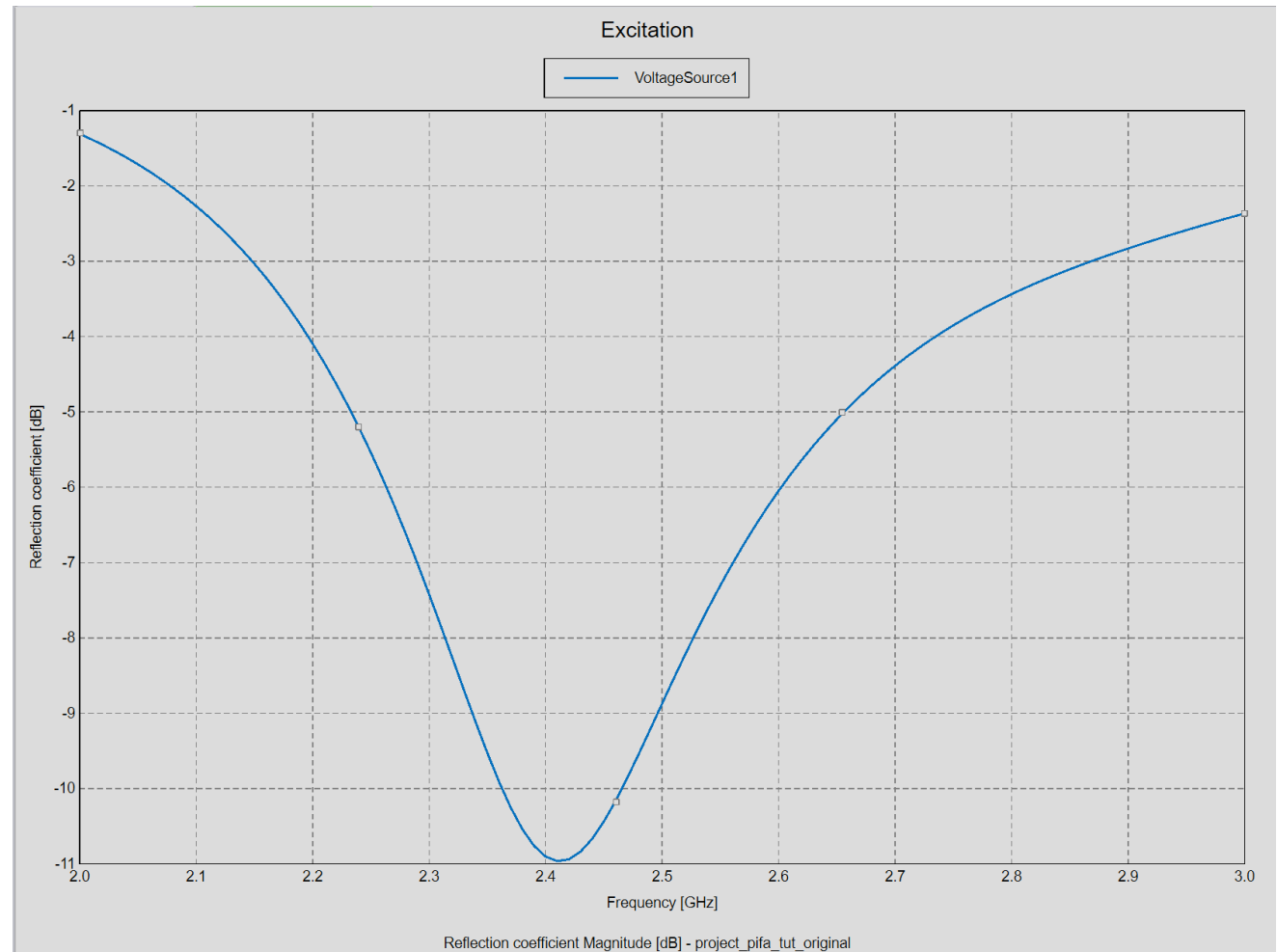
Final Sim Results – 3D Gain Pattern, Underside

- Pattern is quite isotropic, especially in the directions we care about for this application.
- Much lower gain towards chest, which is desirable both for safety and performance



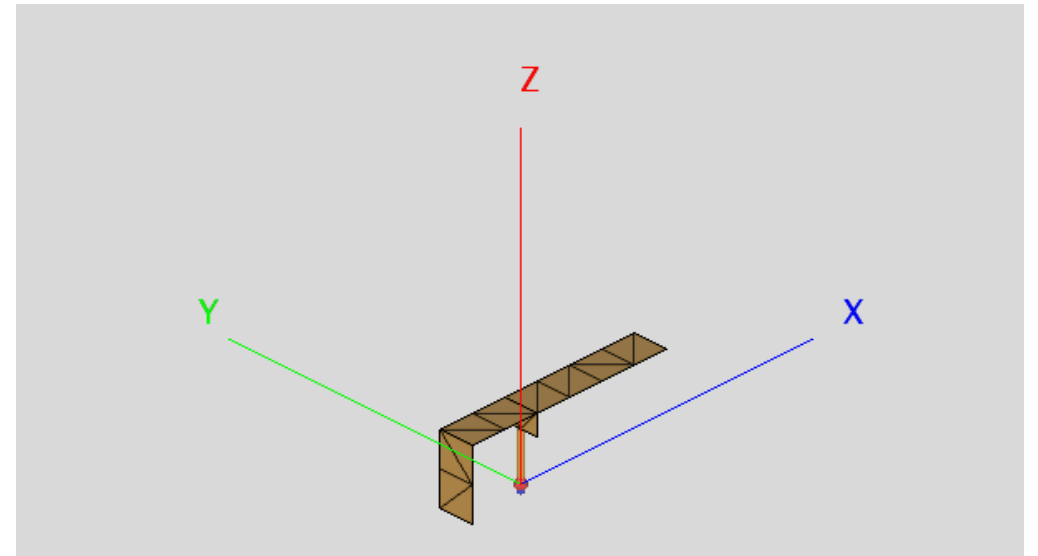
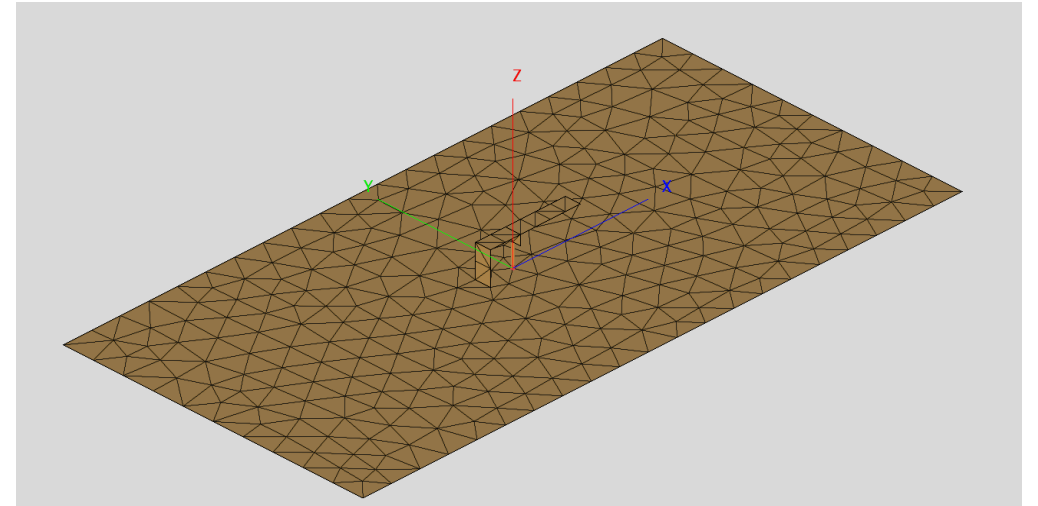
Final Sim Results – Reflection Coefficient

- Resonant at our intended 2.4 GHz, but has enough –10 dB bandwidth to accommodate the 40* 2Mhz channels as well.
- To move resonant frequency swept over L1 and h
- To improve matching ran many sweeps over D and "gap"



Final Values

- $L1 = 30e-3$
- $L2 = W = 5e-3$
- $D = 10e-3$
- $H = 10e-3$
- "Gap" = $7e-3$



Modeling Tissue

IFAC

Calculation of the
Dielectric Properties of Body Tissues
in the frequency range 10 Hz - 100 GHz

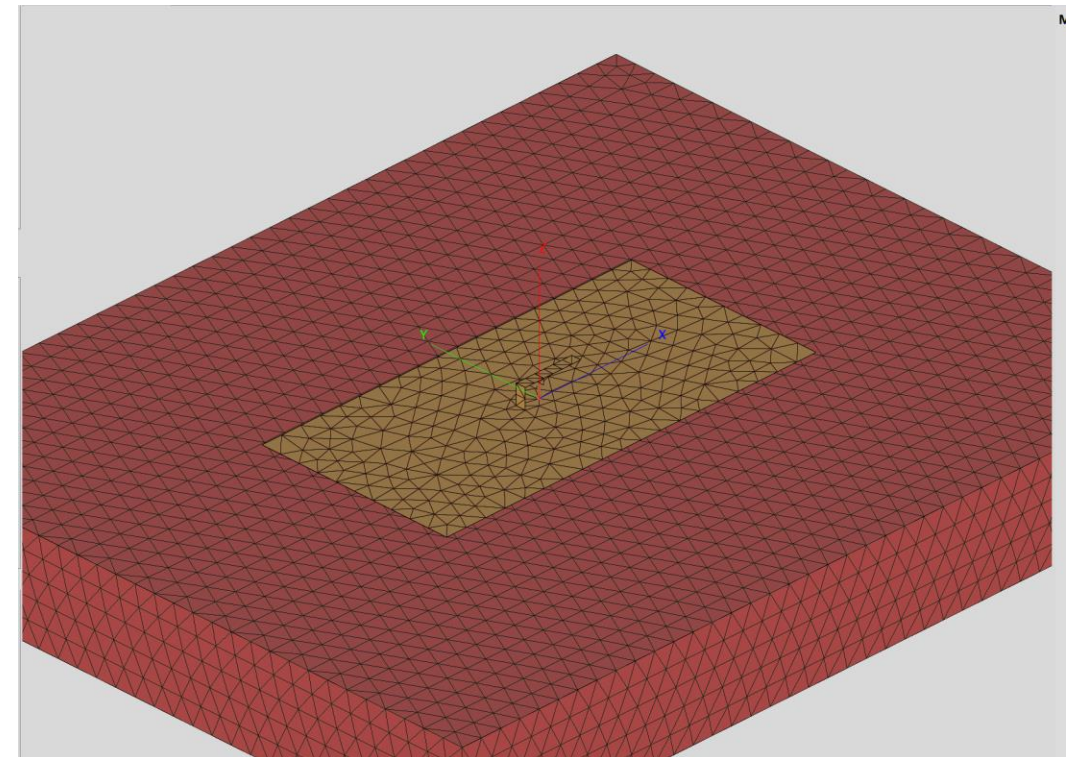
ITALIAN NATIONAL RESEARCH COUNCIL
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SINGLE TISSUE, SINGLE FREQUENCY
Muscle @ 2400000000 Hz

Tissue name	Frequency [Hz]	Conductivity [S/m]	Relative permittivity	Loss tangent	Wavelength [m]	Penetration depth [m]
Muscle	2400000000	1.705	52.791	0.24191	0.017069	0.022785

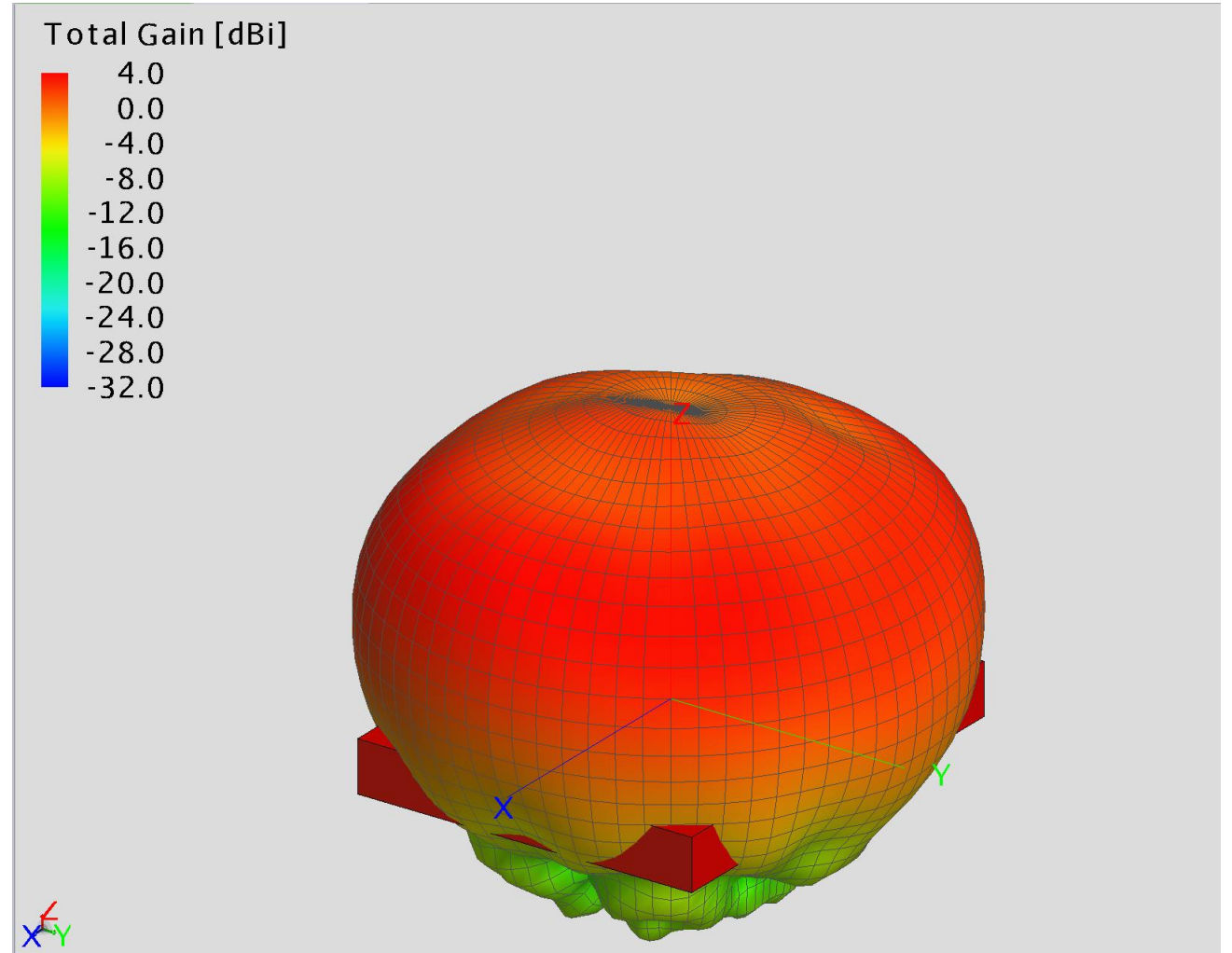
- This antenna will be placed in close proximity to the body, so modeling its electric properties is important.
- There's a database of electric properties of various tissue, so I modeled the chest as a solid block of muscle at 2.4 GHz



Muscle Sim Results

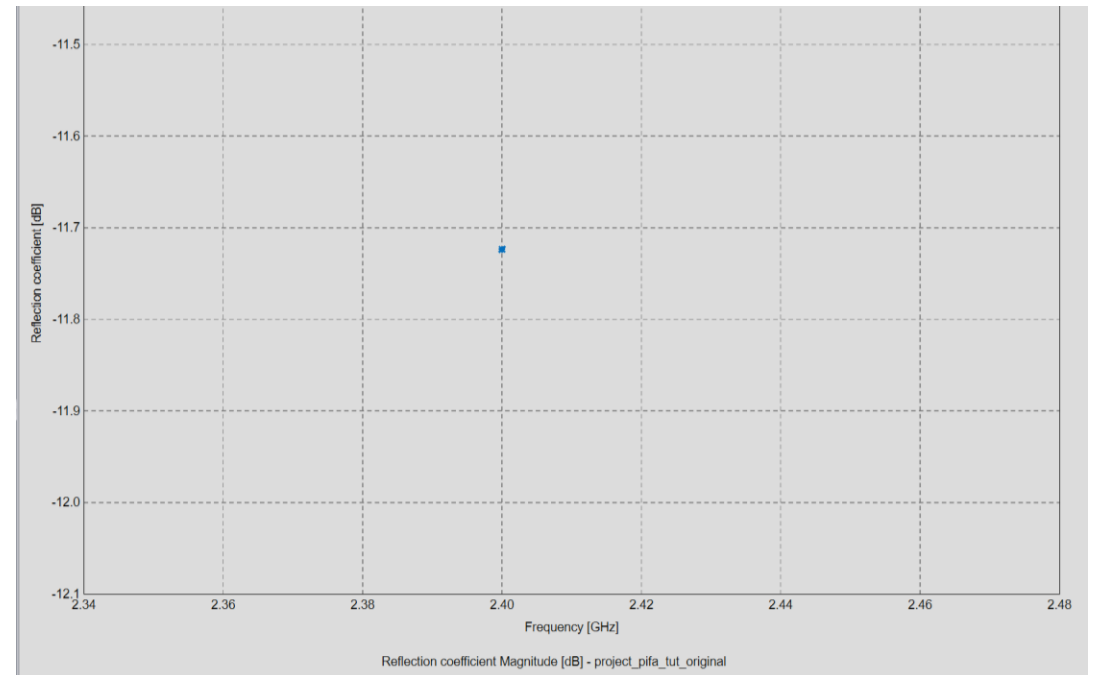
3D Gain Pattern

- Pattern is still quite isotropic in relevant directions
- I would expect this dielectric to reduce performance so it is curious that it appears to not be.



Muscle Sim Results – Reflection Coefficient

- At 2.4 GHz Reflection Coefficient is quite low.
- A sweep was not performed because the model is too large to run many tests.



Sources

PIFA - Planar Inverted-F Antennas

- <https://www.antenna-theory.com/>
- <https://www.sciencedirect.com/topics/engineering/monopole-antenna>
- <https://www.5gtechnologyworld.com/demonstrating-antenna-diversity-part-2-the-pifa/>
- https://www.researchgate.net/figure/Schematic-diagram-of-the-PIFA-antenna_fig1_216605700

BLE-guide

- <https://pubmed.ncbi.nlm.nih.gov/35161734/>
- <https://novelbits.io/bluetooth-low-energy-ble-complete-guide/>
- <https://advanced.onlinelibrary.wiley.com/doi/10.1002/aelm.202400884?af=R>
- <https://www.allaboutcircuits.com/technical-articles/exploring-the-basics-of-bluetooth-low-energy-a-beginners-guide-to-ble/#:~:text=Introduced%20in%202010%20as%20a%20part%20of.they%20used%20up%20too%20much%20battery%20power.>

Electric Properties of Human Tissue

- <https://niremf.ifac.cnr.it/tissprop/htmlclie/htmlclie.php>